

HEALTHGRID AI

Ranchi Healthcare Intelligence Dataset Specification

Tagline: Predict. Prepare. Redistribute. Save.

Purpose: A production-style dataset blueprint for the HealthGrid prototype, combining verified public facility data, clearly labelled simulated operational data, AI-generated predictions, GIS/location intelligence, and chatbot/MCP analytics.

Critical data rule: REAL_VERIFIED, SIMULATED, and AI_GENERATED data must never be presented as the same thing. Missing real fields must remain NULL / Not publicly available rather than being invented.

Ranchi public-source baseline: The official Ranchi district health page lists 14 rural PHC/block areas and reports 328 sub-centres in total, and also lists 22 urban health centres. This PDF is a dataset specification; it does not invent a complete real-time government inventory where the public source does not provide it.

Official source: Ranchi District Administration – Health page: <https://ranchi.nic.in/health/> (verify current data before deployment).

1. DATA LAYERS

Layer	Status	What it contains	Used by
Facility Master	REAL_VERIFIED	Facility identity, name, block, district, coordinates, public address	Public portal / GIS, verified public services/h
Operational	SIMULATED FOR PROTOTYPE	Medicine stock, consumption, patient footfall, beds, staff attendance	Government + ML
AI / ML	AI_GENERATED	Demand forecast, stock-out probability, surge, bed/staff forecasts	Government + Copilot
Optimization	AI_GENERATED	Donor selection, donor safety, transfer quantity, expiry-aware redistribution	Government + Copilot
Location Intelligence	CALCULATED	User GPS, distance, nearby facilities, route/map actions	Public portal + Copilot

2. RANCHI FACILITY MASTER — REAL / VERIFIED

Create one row per verified public facility. Use official government/public sources wherever possible. Do not assume that every administrative block has exactly one PHC. Facility records must be independently represented.

Field	Type	Meaning / rule
facility_id	TEXT	Stable HealthGrid identifier, e.g. RNC-PHC-001; internally generated, not a government ID unless verified.
facility_code	TEXT	Official facility code only when verified.
facility_name	TEXT	Verified public facility name.
facility_type	ENUM	PHC / UPHC / CHC / Urban Health Centre / Sub Centre / Government Hospital etc.
block	TEXT	Verified administrative/health block.
district	TEXT	Ranchi.
state	TEXT	Jharkhand.
latitude	DECIMAL	Verified facility latitude.
longitude	DECIMAL	Verified facility longitude.
address	TEXT	Verified public address.
phone	TEXT	Verified public contact only; otherwise NULL.
email	TEXT	Verified public email only; otherwise NULL.
ownership	TEXT	Government/public ownership information when verified.
beds	INTEGER	Verified public capacity only; otherwise NULL. Do not invent.
doctors	INTEGER	Verified public staffing only; otherwise NULL.
nurses	INTEGER	Verified public staffing only; otherwise NULL.

Field	Type	Meaning / rule
medicines	TEXT/JSON	Publicly documented medicine/pharmacy information only; current stock belongs to operational data.
opening_hours	TEXT/JSON	Verified public operating information.
emergency_available	BOOLEAN	Only when verified.
services	TEXT/JSON	Verified public services.
source	TEXT	Name of authoritative source.
source_url	TEXT	URL of source.
source_verified_date	DATE	Date checked.
data_confidence	ENUM	HIGH / MEDIUM / LOW.
is_verified_real	BOOLEAN	TRUE only for verified public facility data.

3. RANCHI REFERENCE STRUCTURE

The official Ranchi district health page reports the following rural PHC/block areas and sub-centre counts. These are reference administrative/health-structure values, not a claim that each row below is an individual PHC facility.

Area / PHC block	Reported sub-centres
Angara	23
Bero	25
Bundu	17
Burmu	24
Chanho	15
Kanke	34
Lapung	14
Mandar	14
Namkum	25
Ormanjhi	27
Ratu	32
Silli	24
Sonahatu	24
Tamar	30
TOTAL	328

Urban layer: The same official page lists 22 urban health centres. Include those verified facilities in the facility master as appropriate, with facility_type = UPHC / Urban Health Centre where the source supports it.

4. FACILITY SERVICES TABLE

Field	Type
service_id	TEXT
facility_id	TEXT / FK
service_name	TEXT
availability	BOOLEAN / STATUS
source	TEXT
verified_date	DATE
data_type	REAL_VERIFIED

Service examples: OPD, Emergency, Maternal Health, Child Health, Immunization, Pharmacy, Laboratory, Diagnostics, NCD Screening, TB Services, Malaria Services, Family Planning, Antenatal Care, Postnatal Care, Telemedicine, Ambulance. Only mark availability when supported by reliable public data.

5. MEDICINE MASTER

Field	Type	Description
medicine_id	TEXT	Stable medicine identifier.
medicine_name	TEXT	Medicine display name.
generic_name	TEXT	Generic name when applicable.
category	TEXT	Analgesic, antibiotic, ORS, maternal, respiratory, etc.

Field	Type	Description
unit	TEXT	tablet, bottle, vial, strip, etc.
criticality	ENUM	LOW / MEDIUM / HIGH / CRITICAL.
standard_unit	TEXT	Normalized inventory unit.

6. SIMULATED MEDICINE INVENTORY

Field	Type
inventory_id	TEXT
facility_id	FK
medicine_id	FK
date	DATE
opening_stock	INTEGER
received_quantity	INTEGER
consumed_quantity	INTEGER
closing_stock	INTEGER
reorder_level	INTEGER
safety_stock	INTEGER
expiry_date	DATE
days_to_expiry	INTEGER
average_daily_consumption	DECIMAL
days_of_stock	DECIMAL
stock_status	ENUM
data_type	SIMULATED

Formula: $\text{closing_stock} = \text{opening_stock} + \text{received_quantity} - \text{consumed_quantity}$. $\text{days_of_stock} = \text{closing_stock} / \text{average_daily_consumption}$. Statuses: SAFE, WATCH, LOW, CRITICAL, OUT_OF_STOCK.

7. PATIENT FOOTFALL — SIMULATED

Field	Type
record_id	TEXT
facility_id	FK
date	DATE
total_patients	INTEGER
outpatient_cases	INTEGER
emergency_cases	INTEGER
fever_cases	INTEGER
respiratory_cases	INTEGER
maternal_cases	INTEGER
child_cases	INTEGER
other_cases	INTEGER
data_type	SIMULATED

Generate patterned time series: NORMAL, HIGH_DEMAND, SEASONAL_SURGE, and EMERGENCY_SURGE. Patient counts should influence medicine consumption and bed pressure.

8. BED OCCUPANCY — SIMULATED

Field	Type
record_id	TEXT
facility_id	FK
date	DATE
total_beds	INTEGER
occupied_beds	INTEGER
available_beds	INTEGER
occupancy_rate	DECIMAL
data_type	SIMULATED

$\text{available_beds} = \text{total_beds} - \text{occupied_beds}$. $\text{occupancy_rate} = \text{occupied_beds} / \text{total_beds} \times 100$.

9. STAFF ATTENDANCE — SIMULATED

Field	Type
record_id	TEXT
facility_id	FK
date	DATE
doctors_required	INTEGER
doctors_present	INTEGER
nurses_required	INTEGER
nurses_present	INTEGER
other_staff_required	INTEGER
other_staff_present	INTEGER
attendance_rate	DECIMAL
data_type	SIMULATED

10. MEDICINE CONSUMPTION — SIMULATED

Field	Type
record_id	TEXT
facility_id	FK
medicine_id	FK
date	DATE
quantity_consumed	INTEGER
patient_count	INTEGER
consumption_per_patient	DECIMAL
disease_related_demand	DECIMAL
data_type	SIMULATED

Consumption must correlate with patient footfall, disease-category demand, seasonality and recent trends rather than being independent random noise.

11. MEDICINE EXPIRY — SIMULATED

Field	Type
record_id	TEXT
facility_id	FK
medicine_id	FK
batch_number	TEXT
quantity	INTEGER
expiry_date	DATE
days_to_expiry	INTEGER
expiry_risk	ENUM
data_type	SIMULATED

Use FEFO (First Expiry, First Out) logic when deciding whether surplus medicine should be moved to a facility with higher need.

12. AI DEMAND FORECASTS

Field	Type
prediction_id	TEXT
facility_id	FK
medicine_id	FK
prediction_timestamp	TIMESTAMP
forecast_horizon	ENUM: 24H / 48H / 72H
predicted_demand	DECIMAL
confidence	DECIMAL
model_version	TEXT
data_type	AI_GENERATED

13. STOCK-OUT PREDICTIONS

Field	Type
prediction_id	TEXT
facility_id	FK
medicine_id	FK
prediction_time	TIMESTAMP
current_stock	INTEGER
predicted_demand_72h	DECIMAL
days_remaining	DECIMAL
stockout_probability	DECIMAL
expected_stockout_time	TIMESTAMP
risk_level	ENUM
confidence	DECIMAL
model_version	TEXT
data_type	AI_GENERATED

14. PATIENT SURGE PREDICTIONS

Field	Type
facility_id	FK
prediction_date	DATE
current_footfall	INTEGER
predicted_24h	DECIMAL
predicted_48h	DECIMAL
predicted_72h	DECIMAL
surge_probability	DECIMAL
surge_level	ENUM
confidence	DECIMAL
model_version	TEXT
data_type	AI_GENERATED

15. 72-HOUR RISK PREDICTIONS

Field	Type
facility_id	FK
timestamp	TIMESTAMP
risk_now	DECIMAL
risk_24h	DECIMAL
risk_48h	DECIMAL
risk_72h	DECIMAL
medicine_risk	DECIMAL
patient_risk	DECIMAL
bed_risk	DECIMAL
staff_risk	DECIMAL
cascade_risk	DECIMAL

Field	Type
overall_risk_score	DECIMAL
risk_level	ENUM
confidence	DECIMAL
model_version	TEXT
data_type	AI_GENERATED

Suggested risk bands: 0–30 LOW, 31–60 MEDIUM, 61–80 HIGH, 81–100 CRITICAL. Keep weights configurable rather than hardcoded.

16. DATA RELIABILITY

Field	Type
facility_id	FK
data_source	TEXT
last_update	TIMESTAMP
inventory_freshness	DECIMAL
patient_data_freshness	DECIMAL
bed_data_freshness	DECIMAL
staff_data_freshness	DECIMAL
overall_reliability	DECIMAL
confidence_level	ENUM

Real facility identity can have high confidence while simulated operational data is explicitly prototype confidence. Stale operational data should reduce prediction confidence.

17. TRANSFER RECOMMENDATIONS

Field	Type
recommendation_id	TEXT
source_facility	FK
destination_facility	FK
medicine_id	FK
recommended_quantity	INTEGER
source_surplus	INTEGER
destination_need	INTEGER
predicted_demand	DECIMAL
distance_km	DECIMAL
estimated_delivery_time	DECIMAL
source_safety_after_transfer	ENUM
expiry_score	DECIMAL
urgency	ENUM
recommendation_score	DECIMAL
status	ENUM
data_type	AI_GENERATED

Optimization factors: destination urgency, current stock, predicted demand, source surplus, safety stock, distance, delivery time, expiry and donor risk.

18. TRANSFER HISTORY

Field	Type
transfer_id	TEXT
date	DATE
source_facility	FK
destination_facility	FK
medicine_id	FK
quantity	INTEGER
distance_km	DECIMAL
status	ENUM
approved_by	TEXT
dispatched_at	TIMESTAMP
received_at	TIMESTAMP
data_type	SIMULATED / REAL when connected

19. ALERTS

Field	Type
alert_id	TEXT
facility_id	FK
alert_type	ENUM
severity	ENUM

Field	Type
message	TEXT
created_at	TIMESTAMP
status	ENUM
source	TEXT
data_type	AI_GENERATED / SIMULATED

20. CASCADE CRISIS DATA

Field	Type
primary_facility	FK
secondary_facility	FK
trigger	TEXT
predicted_patient_shift	DECIMAL
predicted_bed_pressure	DECIMAL
predicted_medicine_pressure	DECIMAL
cascade_probability	DECIMAL
severity	ENUM
model_version	TEXT
data_type	AI_GENERATED

21. GIS / LOCATION DATA

Field	Type	Purpose
user_latitude	DECIMAL	Browser/Android geolocation; obtained only after user permission.
user_longitude	DECIMAL	Browser/Android geolocation.
facility_latitude	DECIMAL	Verified facility coordinate.
facility_longitude	DECIMAL	Verified facility coordinate.
distance_km	DECIMAL	Calculated geospatial distance.
travel_time	DECIMAL	Optional route-provider estimate.
search_radius_km	DECIMAL	User-selected search radius.
map_action	ENUM	SHOW_ON_MAP / FOCUS_FACILITY / SHOW_ROUTE / SHOW_NEARBY / RISK_LAYER.

Nearest PHC flow: User clicks “Use My Location” → browser/Android requests permission → GPS coordinates → backend geospatial query → verified Ranchi facilities → distance calculation → optional service/open filtering → nearest facilities → map markers + structured chatbot answer.

22. GEOJSON API

GET /api/public/facilities/geojson

Return a GeoJSON FeatureCollection generated from the facilities database. Do not hardcode map markers in the frontend. Each feature should include facility_id, facility_name, facility_type, block, district, address, phone, opening_hours and public services.

23. CHATBOT DATA ANALYST — CORE REQUIREMENT

The HealthGrid chatbot must analyze the same central data platform used by the dashboards. Do not embed the dataset in the prompt. The chatbot should call backend APIs/tools and return structured answers.

Tool	Purpose
get_network_summary	District/network-level summary.
get_facility	Facility profile.
search_facilities	Search by name/type/block.
get_facilities_near_location	GIS nearest-facility query.
get_inventory_status	Medicine inventory.
get_low_stock_medicines	Shortage monitoring.
get_medicine_consumption	Consumption trends.
get_patient_footfall	Historical footfall.
get_patient_forecast	24/48/72-hour forecast.
get_patient_surge_prediction	Surge probability.
get_bed_availability	Current public/authorized bed data.
get_bed_forecast	Capacity forecast.
get_staff_status	Authorized staff status.
get_active_alerts	Current alerts.
get_phc_risk	Facility risk explanation.
get_high_risk_phcs	Ranked risk facilities.
get_72_hour_risk	Time-horizon risk.
find_surplus_medicine	Potential donor facilities.
check_donor_safety	Prevent harmful transfers.
calculate_distance	GIS distance.
recommend_medicine_transfer	Optimization recommendation.
get_transfer_status	Transfer lifecycle.
simulate_emergency	What-if simulation.
get_data_reliability	Data freshness/confidence.

24. CHATBOT QUESTIONS TO SUPPORT

- Which PHCs in Ranchi are at highest risk?
- Which medicines are likely to run out in the next 72 hours?
- Show critical PHCs on the map.
- Find the nearest PHC to me.
- What government facilities are within 5 km?
- Which nearby facility provides maternal health services?
- Which PHC has the highest patient surge?
- Which facilities have the lowest bed availability?
- Where is Paracetamol running low?
- Which PHCs have surplus Paracetamol?
- Why is PHC X at high risk?
- Why did HealthGrid recommend this transfer?
- Compare PHC A and PHC B.
- What happens if patient footfall increases by 30%?
- Which block has the highest patient load?
- Which medicine has the highest consumption?

25. STRUCTURED CHATBOT RESPONSE FORMAT

Example analytical answer:

■ 72-HOUR MEDICINE RISK — RANCHI

3 facilities require attention.

1. Facility A

Medicine: Paracetamol

Stock-out probability: 87%

Expected shortage: ~2 days

Risk: CRITICAL

RECOMMENDED ACTION

Potential donor: Facility D

Donor safety: SAFE

Distance: 18 km

Recommended quantity: 500 units

MAP

Show critical facilities + donor + transfer route.

DATA STATUS

Facility/location: REAL VERIFIED

Operations: SIMULATED FOR PROTOTYPE

Predictions: AI GENERATED

26. CHATBOT + MAP ACTION CONTRACT

The chatbot should return an internal structured UI action, for example:

```
{
  "answer": "...",
  "map_action": {
    "type": "SHOW_ON_MAP",
    "facility_ids": ["RNC-PHC-001", "RNC-PHC-008"]
  }
}
```

The frontend uses facility IDs to focus the map. Do not expose raw JSON to normal users.

27. ROLE-BASED CHATBOT ACCESS

Role	Allowed information
PUBLIC	Public facility identity, location, services, public contact and public operating information.
PHC_STAFF	Own facility operational data, authorized inventory/beds/staff, requests, alerts and authorized coordination.
DISTRICT_OFFICIAL	District-wide risk, inventory, predictions, transfers and intelligence.
STATE_OFFICIAL	State-level intelligence.
NATIONAL_ADMIN	Network-wide intelligence.
SYSTEM_ADMIN	System administration and audit controls.

28. DATASET FILES TO GENERATE

- ranchi_facilities.csv
- ranchi_blocks.csv
- ranchi_facility_services.csv
- medicine_master.csv
- medicine_inventory_simulated.csv
- medicine_consumption_simulated.csv
- medicine_expiry_simulated.csv
- patient_footfall_simulated.csv
- bed_occupancy_simulated.csv
- staff_attendance_simulated.csv
- medicine_transfers_simulated.csv
- alerts_simulated.csv
- ai_prediction_features.csv
- demand_forecasts.csv
- stockout_predictions.csv
- patient_surge_predictions.csv

- bed_forecasts.csv
- staff_predictions.csv
- risk_predictions.csv
- cascade_predictions.csv
- transfer_recommendations.csv
- data_sources.csv

29. SYNTHETIC DATA SCENARIOS

Scenario	Pattern
NORMAL	Stable patients, stock, beds and staffing.
MEDICINE_SHORTAGE	Demand rising, stock falling, shortage predicted before zero.
PATIENT_SURGE	Football spike drives medicine and bed pressure.
BED_CRISIS	Occupancy becomes critical.
STAFF_SHORTAGE	Doctor/nurse availability falls.
CASCADE_CRISIS	One facility's failure shifts pressure to another.
EXPIRY_RISK	Surplus medicine approaches expiry and can be safely redistributed.
DONOR_AVAILABLE	Source facility has surplus while remaining above safety stock.
EMERGENCY	Multiple resources become stressed simultaneously.

30. END-TO-END HEALTHGRID LOOP

```

REAL VERIFIED RANCHI FACILITY DATA
→ GIS
→ SIMULATED OPERATIONAL SIGNALS
→ AI/ML FORECASTS
→ RISK ENGINE
→ RESOURCE OPTIMIZATION
→ CHATBOT EXPLANATION
→ MAP VISUALIZATION
→ GOVERNMENT APPROVAL
→ PHC ACTION
→ DATA UPDATE
→ AI RE-CALCULATION
→ HEALTHGRID

```

31. RECOMMENDED PROJECT FILES

DATASET.md – data sources, field definitions, real/simulated/AI labels.
 DATABASE.md – schema and relationships.
 ML.md – feature engineering, models, evaluation and confidence.
 GIS.md – geolocation, distance, GeoJSON and map integration.
 CHATBOT.md – tools, structured responses, role-based access and map actions.
 MCP.md – MCP tools and backend integration.
 API.md – endpoint contracts.
 DEMO.md – hackathon demonstration scenarios.

32. DATA GOVERNANCE CHECKLIST

- Never invent a real facility name, address, phone, coordinate, bed count, doctor count or nurse count.
- Preserve source and verification date for real facility fields.
- Keep simulated operational data clearly labelled.
- Keep AI outputs clearly labelled with model version and timestamp.
- Do not expose sensitive internal operational data to public users.
- Use RBAC for Government, PHC and Public access.
- Do not use the public chatbot for diagnosis or prescriptions.
- Do not hardcode KPIs or map markers.

- All portals and chatbot must use the same central APIs/database.
- Keep an audit trail for resource recommendations and approvals.

FINAL POSITIONING

HealthGrid is not just a dashboard or chatbot. It is a connected healthcare intelligence network that uses verified Ranchi facility data as the geographic foundation, prototype operational signals as a controlled simulation layer, AI/ML for prediction, GIS for location intelligence, optimization for resource redistribution, and an AI Copilot that can analyze the same data and visualize answers directly on the map.

Source note: Ranchi District Administration – Health: <https://ranchi.nic.in/health/> . Verify current public data and field-level availability before production deployment.